

Systematic Derivation of the Three $\frac{1}{2}\langle 111 \rangle \rightarrow \langle 100 \rangle$ Loop-Conversion Kernels

Thermodynamic rotation, collision-induced junction formation, and absorption growth

Technical derivation for the RadCluster formulation

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Abstract

This note derives, step by step, the three rate kernels used to transfer self-interstitial content between mobile $\frac{1}{2}a\langle 111 \rangle$ loops and sessile $a\langle 100 \rangle$ loops in RadCluster. The unary kernel is constructed by combining the anisotropic-elastic free-energy difference of Dudarev et al. (2008) with transition-state kinetics and a one-way net-rate closure. The junction kernel is obtained by multiplying the binary encounter kernel by (i) a size-and-orientation compatibility factor and (ii) the probability that the metastable $\frac{1}{2}\langle 110 \rangle$ intermediate of Marian et al. (2002) reaches $\langle 100 \rangle$ before reverting. The absorption kernel follows from the same diffusion-limited encounter problem, but only the $\frac{1}{2}\langle 111 \rangle$ partner is mobile and the pre-existing $\langle 100 \rangle$ loop templates the product character. Every empirical or calibrated closure is distinguished from quantities taken from elasticity, atomistics, or geometry. Numerical checks expose a parameter inconsistency in the supplied draft and give the corrected barrier law.

1 Reaction network, state variables, and units

Let c_n^{111} and c_m^{100} denote the atomic fractions (number per lattice site) of interstitial clusters containing n and m SIAs. If number densities are used instead, write $C_n = c_n/\Omega$, where Ω is the atomic volume. The three admissible reactions are



Here 111_n abbreviates a $\frac{1}{2}a\langle 111 \rangle$ loop and 100_m an $a\langle 100 \rangle$ loop. The 100 population is defined only for $m \geq n_{\text{loop}}^{\min}$ and is sessile, $D_m^{100} = 0$. The 111 population is mobile only for $n \leq i_{\text{mobile}}$.

For atomic-fraction variables, a unary kernel has units s^{-1} and a binary kernel also has units s^{-1} :

$$\dot{c}_a = -\Gamma_a c_a, \quad \text{quad} \dot{c}_a = -K_{ab} c_a c_b. \quad (4)$$

The corresponding volumetric binary coefficient is $k_{ab}^{(V)} = \Omega K_{ab}$, with units $\text{m}^3 \text{s}^{-1}$, because $\dot{C}_a = -k_{ab}^{(V)} C_a C_b$.

Table 1: Logical factorization of the three kernels. Factors labeled “closure” are not uniquely determined by the cited microscopic theory.

Kernel	Attempt or encounter	Selection/success	Admissibility
$\Gamma_{\text{uni}}(n, T)$	transition-state attempt rate	Dudarev driving-force gate (closure)	loop floor; $\Delta F > 0$
$K_{nn'}^{\text{junc}}$	111–111 encounter kernel	size/orientation compatibility and Marian completion probability (closures)	both partners mobile; size floor
K_{mn}^{abs}	mobile 111 encounter with sessile 100	templated completion probability (closure)	$m \geq n_{\text{loop}}^{\min}$; $n \leq i_{\text{mobile}}$; $m+n \leq I$

2 Common geometry and loop free energy

2.1 Area, perimeter, and size scaling

Insertion of n atomic volumes produces a platelet volume $A_X b_X = n\Omega$. Thus, for character $X \in \{111, 100\}$,

$$A_X(n) = \frac{n\Omega}{b_X}, \quad b_{111} = \frac{\sqrt{3}}{2}a, \quad b_{100} = a. \quad (5)$$

For the hexagonal 111 and square 100 shapes used in RadCluster,

$$P_{111}(n) = 6\sqrt{\frac{2A_{111}(n)}{3\sqrt{3}}}, \quad P_{100}(n) = 4\sqrt{A_{100}(n)}. \quad (6)$$

Substitution of Eq. (5) into Eq. (6) gives

$$\frac{P_{111}(n)}{b_{111}} = \underbrace{\frac{6}{b_{111}}\sqrt{\frac{2\Omega}{3\sqrt{3}b_{111}}}}_{C_P} \sqrt{n}, \quad (7)$$

$$\frac{P_{100}(n)}{b_{100}} = \frac{4}{b_{100}}\sqrt{\frac{\Omega}{b_{100}}}\sqrt{n}. \quad (8)$$

Hence all perimeter-proportional energetic terms scale as $\sqrt{n}$.

2.2 Dudarev anisotropic-elastic free energy

The loop formation free energy is represented by

$$E_l^X(n, T) = P_X(n) \left\{ \hat{F}_X(T) \ln \left[\frac{4R_X^*(n)}{e\delta} \right] + F_\delta^X + F_c^X \right\}, \quad (9)$$

where $R_X^* = \sqrt{A_X/\pi}$ is an equal-area radius, δ is the core cutoff, $\hat{F}_X$ is the anisotropic prelogarithmic line-energy factor, and $F_\delta^X + F_c^X$ represents core traction plus nonlinear core energy (Dudarev et al., 2008). The conversion driving force at fixed SIA content is

$$\Delta F(n, T) = E_l^{111}(n, T) - E_l^{100}(n, T). \quad (10)$$

The sign convention is important: $\Delta F > 0$ means that the 100 product is thermodynamically lower in free energy.

For a pure-edge 001[100] dislocation, anisotropic elasticity gives

$$\hat{F}_{001}([100], T) = \frac{a^2}{4\pi}(c_{11} + c_{12}) \left[\frac{c_{44}(c_{11} - c_{12})}{c_{11}(c_{11} + c_{12} + 2c_{44})} \right]^{1/2}. \quad (11)$$

Near the bcc–fcc transformation, the tetragonal shear modulus obeys the empirical softening law

$$c'(T) = \frac{c_{11} - c_{12}}{2} \simeq c'_0(1 - T/T_c)^{1/2}, \quad (12)$$

so Eq. (11) implies

$$\hat{F}_{100}(T) = \hat{F}_{100}^0(1 - T/T_c)^{1/4}. \quad (13)$$

This is the physical source of the strong temperature dependence: the 100 elastic line energy softens, while the relevant 111 factors remain finite (Dudarev et al., 2008).

The fitted zero-temperature coefficient $\hat{F}_{100}^0$ is obtained from one crossover condition, not independently prescribed. Define

$$L_X(n) = \ln \left[\frac{4R_X^*(n)}{e\delta} \right], \quad Q(T) = (1 - T/T_c)^{1/4}. \quad (14)$$

Imposing $\Delta F(n_{\text{ref}}, T^*) = 0$ in Eq. (10) yields

$$\boxed{\hat{F}_{100}^0 = \frac{P_{111}}{P_{100}} \left[\hat{F}_{111} L_{111} + F_{\delta}^{111} + F_c^{111} \right] - (F_{\delta}^{100} + F_c^{100})} \quad \text{at } n = n_{\text{ref}}. \quad (15)$$

Equation (15) documents exactly which datum is calibrated and prevents $\hat{F}_{100}^0$ and T^* from being counted as independent inputs.

3 Kernel 1: unary $111_n \rightarrow 100_n$ rotation

3.1 Step 1: transition-state attempt rate

For a 111 loop trapped in a local free-energy minimum, transition-state theory gives

$$k_+(n, T) = \nu_0 \exp \left[-\frac{E_a(n)}{k_B T} \right], \quad (16)$$

where $E_a = F^\ddagger - F_{111}$ is the forward activation barrier and ν_0 is an attempt frequency. The available atomistic literature motivates a thermally activated reorientation pathway, but does not supply a unique barrier for every loop size. The RadCluster closure assumes that the activated portion of the loop line grows with perimeter:

$$E_a(n) = E_{a0} + \gamma_a \frac{P_{111}(n)}{b_{111}}. \quad (17)$$

Here P_{111}/b_{111} is the number of Burgers-vector-length line segments and γ_a is an effective barrier per segment.

3.2 Step 2: thermodynamic direction and the net-rate factor

For an explicitly reversible pair of states, local detailed balance requires

$$\frac{k_+(n, T)}{k_-(n, T)} = \exp \left[\frac{\Delta F(n, T)}{k_B T} \right]. \quad (18)$$

Choosing $k_- = k_+ \exp[-\Delta F/(k_B T)]$ gives the forward-minus-reverse coefficient

$$k_+ - k_- = k_+ \left[1 - \exp \left(-\frac{\Delta F}{k_B T} \right) \right]. \quad (19)$$

RadCluster uses Eq. (19) as an effective irreversible rate only where the forward reaction is downhill. Defining $[x]_+ = \max(0, x)$ therefore gives

$$\boxed{\Gamma_{\text{uni}}(n, T) = \nu_0 \exp \left[-\frac{E_a(n)}{k_B T} \right] \left[1 - \exp \left[-\frac{\Delta F(n, T)}{k_B T} \right] \right]_+ \Theta(n - n_{\text{loop}}^{\min})}. \quad (20)$$

The bracket has the correct limiting behavior:

$$\left[1 - e^{-\Delta F/k_B T} \right]_+ \simeq \begin{cases} 0, & \Delta F \leq 0, \\ \Delta F/(k_B T), & 0 < \Delta F \ll k_B T, \\ 1, & \Delta F \gg k_B T. \end{cases} \quad (21)$$

This construction is best described as a *one-way net-rate closure consistent with the sign of the Dudarev driving force*. It is not a fully reversible detailed-balance model, because the actual term $+k_- c_n^{100}$ is not retained. A reversible alternative would be

$$\dot{c}_n^{111} = -k_+ c_n^{111} + k_- c_n^{100}, \quad \dot{c}_n^{100} = +k_+ c_n^{111} - k_- c_n^{100}. \quad (22)$$

3.3 Step 3: population equation and conservation

The implemented unary contribution is

$$\dot{c}_n^{111}|_{\text{uni}} = -\Gamma_{\text{uni}}c_n^{111}, \quad \dot{c}_n^{100}|_{\text{uni}} = +\Gamma_{\text{uni}}c_n^{111}. \quad (23)$$

Therefore

$$\left. \frac{d}{dt} \sum_n n(c_n^{111} + c_n^{100}) \right|_{\text{uni}} = 0, \quad (24)$$

and loop number is also conserved by this channel.

3.4 Numerical reduction of the barrier

Using $a = 2.867 \text{ \AA}$, $\Omega = 11.80 \text{ \AA}^3$, and Eq. (7) gives

$$b_{111} = 2.483 \text{ \AA}, \quad C_P = 3.26835, \quad qquad \frac{P_{111}(n)}{b_{111}} = 3.26835\sqrt{n}. \quad (25)$$

With the supplied table values $E_{a0} = 2.10 \text{ eV}$ and $\gamma_a = 0.020 \text{ eV}$,

$$\boxed{E_a(n) = 2.10 + 0.06537\sqrt{n} \text{ eV}.} \quad (26)$$

Consistency correction. The expression $E_a = 1.80 + 0.098\sqrt{n} \text{ eV}$ in the supplied prose corresponds to the earlier defaults $E_{a0} = 1.8 \text{ eV}$ and $\gamma_a = 0.03 \text{ eV}$ found in the current repository, not to the supplied parameter table. With that table, Eq. (26) is the correct reduction. Similarly, if $\Delta F > 0$ only for $1 \leq n \leq 23$ and $n_{\text{loop}}^{\min} = 4$, the actual support of Eq. (20) is $4 \leq n \leq 23$.

4 Kernel 2: $111_n + 111_{n'} \rightarrow 100_{n+n'}$ junction

4.1 Step 1: diffusion-limited encounter rate

For two three-dimensionally diffusing objects, introduce the relative coordinate $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$ and relative diffusivity $D_{\text{rel}} = D_n + D_{n'}$. Outside an absorbing encounter surface of radius $R_c = R_n^c + R_{n'}^c$, the steady pair concentration obeys

$$D_{\text{rel}}\nabla^2 C = 0, \quad C(R_c) = 0, \quad C(r \rightarrow \infty) = C_\infty. \quad (27)$$

The spherical solution is $C(r) = C_\infty(1 - R_c/r)$. Integrating the inward flux at R_c gives the Smoluchowski coefficient

$$k_{nn'}^{(V)} = \int_{S_c} D_{\text{rel}}\nabla C \cdot d\mathbf{S}/C_\infty = 4\pi R_c(D_n + D_{n'}). \quad (28)$$

To reproduce the radius convention used in RadCluster, write

$$R_n^c = 2\Omega^{1/3}\xi_n, \quad \xi_n = \left(\frac{3n}{8\pi}\right)^{1/3}. \quad (29)$$

Dividing Eq. (28) by Ω converts the volumetric coefficient to the atomic-fraction kernel of Eq. (4):

$$\boxed{\mathcal{K}_{n,n'}^{ii} = \frac{8\pi}{\Omega^{2/3}}(\xi_n + \xi_{n'})(D_n^{111} + D_{n'}^{111}).} \quad (30)$$

If D^{111} is the isotropic-equivalent diffusivity of a 1D/3D glider, then Eq. (30) is an effective-medium closure. Pure 1D and mixed-dimensional encounters generally require different first-passage coefficients; analytical and kinetic Monte Carlo treatments are discussed by Huang and Marian (2019) and the effect on ferritic-iron cluster dynamics by Kohnert and Wirth (2015).

4.2 Step 2: branching at encounter

The Burgers-vector reaction identified by Marian et al. (2002) is

$$\frac{1}{2}[111] + \frac{1}{2}[\bar{1}\bar{1}\bar{1}] \longrightarrow [100]. \quad (31)$$

Not every encounter has the required Burgers-vector combination, geometry, impact parameter, or relative size. The binary collision kernel is therefore split without changing the total encounter budget:

$$K_{nn'}^{111 \rightarrow 100} = \varphi_{\text{junc}}(n, n', T) \mathcal{K}_{n, n'}^{ii}, \quad (32)$$

$$K_{nn'}^{111 \rightarrow 111} = [1 - \varphi_{\text{junc}}(n, n', T)] \mathcal{K}_{n, n'}^{ii}, \quad (33)$$

$$K_{nn'}^{111 \rightarrow 100} + K_{nn'}^{111 \rightarrow 111} = \mathcal{K}_{n, n'}^{ii}. \quad (34)$$

4.3 Step 3: completion probability of the metastable intermediate

The Marian pathway contains a metastable $\frac{1}{2}\langle 110 \rangle$ configuration. From that state, let completion and reversal be independent activated events:

$$k_2 = \nu_2 e^{-\Delta H_2 / k_B T}, \quad k_{\text{rev}} = \nu_r e^{-\Delta H_{\text{rev}} / k_B T}. \quad (35)$$

For two independent exponential waiting times, the probability that completion occurs first is

$$P_{\text{succ}}(T) = \int_0^\infty \underbrace{k_2 e^{-k_2 t}}_{\text{completion at } t} \underbrace{e^{-k_{\text{rev}} t}}_{\text{no reversal before } t} dt \quad (36)$$

$$= \frac{k_2}{k_2 + k_{\text{rev}}} \quad (37)$$

$$= \left[1 + \frac{\nu_r}{\nu_2} \exp\left(\frac{\Delta H_2 - \Delta H_{\text{rev}}}{k_B T}\right) \right]^{-1}. \quad (38)$$

Assuming equal attempt frequencies, $\nu_2 = \nu_r$, gives the implemented gate

$$\boxed{P_{\text{succ}}(T) = \left[1 + \exp\left(\frac{\Delta H_2 - \Delta H_{\text{rev}}}{k_B T}\right) \right]^{-1}}. \quad (39)$$

Only the barrier difference is identifiable from this scalar gate. Separate barriers would become identifiable only if an absolute residence time or total escape rate from the intermediate were also used.

4.4 Step 4: size compatibility and final junction kernel

The unresolved geometrical compatibility is represented by a log-size Gaussian:

$$g(n, n') = \exp\left[-\frac{\ln^2(n/n')}{2\sigma_s^2}\right]. \quad (40)$$

It is symmetric, $g(n, n') = g(n', n)$; it peaks at equal size; and $g = e^{-1/2}$ at $n/n' = e^{\pm\sigma_s}$. Combining compatibility, completion, and the minimum size gives

$$\boxed{\varphi_{\text{junc}} = P_{\text{succ}}(T) \varphi_{\text{max}} g(n, n') \Theta[\min(n, n') - n_j^{\text{min, eff}}]}, \quad (41)$$

where

$$n_j^{\text{min, eff}} = \max\{2, \min[n_j^{\text{min}}, \lceil f_j i_{\text{mobile}} \rceil]\}. \quad (42)$$

Equation (42) is a numerical admissibility rule: it prevents a calibrated junction channel from becoming identically inaccessible when $i_{\text{mobile}} < n_j^{\text{min}}$. It is not derived from Marian atomistics and should be reported as a model regularization. Combining Eqs. (30), (32), and (41) gives the requested second kernel.

4.5 Step 5: mass-action equations and factors of two

For a symmetric kernel, the product gain at size p is

$$\dot{c}_p^{100}|_{\text{junc}} = \frac{1}{2} \sum_{n=1}^{p-1} K_{n,p-n}^{111 \rightarrow 100} c_n^{111} c_{p-n}^{111}, \quad (43)$$

where $1/2$ prevents double counting $(n, p-n)$ and $(p-n, n)$. The reactant loss is

$$\dot{c}_n^{111}|_{\text{junc}} = -c_n^{111} \sum_{n'} K_{nn'}^{111 \rightarrow 100} c_{n'}^{111}. \quad (44)$$

Then, after relabeling the double sums,

$$\left. \frac{d}{dt} \left(\sum_n n c_n^{111} + \sum_p p c_p^{100} \right) \right|_{\text{junc}} = 0. \quad (45)$$

Loop number is not conserved: every junction converts two loops into one.

5 Kernel 3: $100_m + 111_n \rightarrow 100_{m+n}$ absorption

5.1 Step 1: relative diffusivity and encounter kernel

The general cross-character Smoluchowski kernel in atomic-fraction units is

$$\mathcal{K}_{mn}^{100,111} = \frac{8\pi}{\Omega^{2/3}} (\xi_m + \xi_n) (D_m^{100} + D_n^{111}). \quad (46)$$

Because the host 100 loop is declared sessile,

$$D_m^{100} = 0 \implies \mathcal{K}_{mn}^{100,111} = \frac{8\pi}{\Omega^{2/3}} (\xi_m + \xi_n) D_n^{111}. \quad (47)$$

The asymmetry therefore arises from mobility and product identity, not from discarding the host size: m remains in the capture radius.

5.2 Step 2: a distinct templated-completion gate

The captured 111 cluster must be incorporated into the character of the much larger 100 host. The same competing-hazard argument as Eqs. (36)–(38), but with an absorption-specific forward barrier and prefactor, gives the closure

$$P_{\text{succ}}^{\text{abs}}(T) = A_{\text{abs}} \frac{e^{-\Delta H_2^{\text{abs}}/k_B T}}{e^{-\Delta H_2^{\text{abs}}/k_B T} + e^{-\Delta H_{\text{rev}}/k_B T}} = \frac{A_{\text{abs}}}{1 + e^{(\Delta H_2^{\text{abs}} - \Delta H_{\text{rev}})/k_B T}}. \quad (48)$$

This is not a literal probability unless parameters ensure $0 \leq P_{\text{succ}}^{\text{abs}} \leq 1$. With $\Delta H_2^{\text{abs}} > \Delta H_{\text{rev}}$, the high-temperature limit is $A_{\text{abs}}/2$, so $A_{\text{abs}} \leq 2$ preserves the probability interpretation. More generally, Eq. (48) should be called an *effective success multiplier* or clipped at unity.

The physical reason for allowing $\Delta H_2^{\text{abs}} \neq \Delta H_2$ is that junction formation creates a new 100 nucleus from two comparable 111 loops, whereas absorption is templated by an existing 100 loop. This distinction is a RadCluster hypothesis motivated by the Marian mechanism, not a barrier directly reported for this exact coarse-grained reaction.

5.3 Step 3: final absorption kernel and admissibility

Multiplying Eqs. (47) and (48) gives

$$K_{mn}^{\text{abs}} = P_{\text{succ}}^{\text{abs}}(T) \frac{8\pi}{\Omega^{2/3}} (\xi_m + \xi_n) D_n^{111}, \quad (49)$$

subject to

$$m \geq n_{\text{loop}}^{\min}, \quad n \leq i_{\text{mobile}}, \quad m + n \leq I. \quad (50)$$

If the implementation contains a separate multiplier `absorb_boost_100`, it multiplies the right-hand side of Eq. (49) and should be listed independently from $P_{\text{succ}}^{\text{abs}}$.

5.4 Step 4: master-equation terms and conservation

The mobile-partner loss is

$$\dot{c}_n^{111}|_{\text{abs}} = -c_n^{111} \sum_m K_{mn}^{\text{abs}} c_m^{100}. \quad (51)$$

For the 100 ladder, absorption moves a loop from m to $m + n$:

$$\dot{c}_p^{100}|_{\text{abs}} = \sum_{n=1}^{p-n_{\text{loop}}^{\min}} K_{p-n,n}^{\text{abs}} c_{p-n}^{100} c_n^{111} - c_p^{100} \sum_n K_{pn}^{\text{abs}} c_n^{111}. \quad (52)$$

Multiplying Eqs. (51) and (52) by cluster size and summing shows

$$\left. \frac{d}{dt} \left(\sum_n n c_n^{111} + \sum_m m c_m^{100} \right) \right|_{\text{abs}} = 0. \quad (53)$$

The bottom-boundary rule—a 100 loop shrinking below $n_{\text{loop}}^{\min}$ returns to the 111 population—is required to retain this conservation when other growth, shrinkage, and emission edges operate.

6 Numerical evaluation of the two success gates

With $\Delta H_2 = 0.44$ eV, $\Delta H_{\text{rev}} = 0.30$ eV, $\Delta H_2^{\text{abs}} = 0.36$ eV, $A_{\text{abs}} = 2$, and $k_B = 8.617333262 \times 10^{-5}$ eV K⁻¹, Eqs. (39) and (48) give Table 2.

Table 2: Recalculated completion factors for the supplied calibrated barriers.

T (°C)	T (K)	P_{succ}	$P_{\text{succ}}^{\text{abs}}$
300	573.15	0.0555	0.4577
330	603.15	0.0634	0.4794
400	673.15	0.0822	0.5245

The absorption multiplier is 8.25, 7.57, and 6.38 times the junction completion probability at 300, 330, and 400 °C. The ratio decreases with temperature, although both gates increase. The actual junction yield is smaller still by $\varphi_{\text{max}} g(n, n')$.

7 Complete parameter provenance and consistency table

8 Compact final forms and interpretation

The three derived kernels are

$$\Gamma_{\text{uni}}(n, T) = \nu_0 e^{-[E_{a0} + \gamma_a P_{111}(n)/b_{111}]/k_B T} \left[1 - e^{-\Delta F(n, T)/k_B T} \right]_+ \Theta(n - n_{\text{loop}}^{\min}), \quad (54)$$

$$K_{nn'}^{\text{junc}} = P_{\text{succ}}(T) \varphi_{\text{max}} e^{-\ln^2(n/n')/(2\sigma_s^2)} \Theta[\min(n, n') - n_j^{\text{min,eff}}] \frac{8\pi}{\Omega^{2/3}} (\xi_n + \xi_{n'}) (D_n^{111} + D_{n'}^{111}), \quad (55)$$

$$K_{mn}^{\text{abs}} = P_{\text{succ}}^{\text{abs}}(T) \frac{8\pi}{\Omega^{2/3}} (\xi_m + \xi_n) D_n^{111}. \quad (56)$$

Their physical roles are distinct:

- Equation (54) transfers an existing loop between character states. Dudarev theory supplies ΔF ; the barrier law and one-way net-rate interpretation are mesoscale closures.
- Equation (55) creates a new 100 loop through an encounter between two mobile 111 loops. Marian atomistics motivates the reaction path; the log-Gaussian compatibility function and effective barriers coarse-grain unresolved collision variables.
- Equation (56) grows an existing sessile 100 loop by capture of a mobile 111 loop. Its separate completion gate represents templating by the host and should not be conflated with junction nucleation.

Table 3: Inputs to Eqs. (20), (32), and (49). “Model/calibration” means that the numerical value is not directly supplied by the cited atomistic or elastic theory.

Quantity	Symbol	Value	Basis/provenance
Attempt frequency	ν_0	10^{13} s^{-1}	phonon-scale prefactor; model input
Unary barrier intercept	E_{a0}	2.10 eV	supplied calibration; differs from repository default 1.8 eV
Barrier per line segment	γ_a	0.020 eV	supplied calibration; differs from repository default 0.03 eV
Loop onset	$n_{\text{loop}}^{\min}$	4	population-definition/model input
Peak junction efficiency	$\varphi_{\max}$	0.10	unresolved orientation/geometry; calibration
Log-size width	σ_s	0.35	size-compatibility closure; calibration
Nominal junction floor	$n_j^{\min}$	30	model/calibration
Mobility fraction	f_j	0.60	numerical admissibility regularization
Junction completion barrier	ΔH_2	0.44 eV	calibrated effective barrier inspired by the Marian intermediate
Reverse barrier	ΔH_{rev}	0.30 eV	calibrated/effective Marian-pathway barrier
Absorption completion barrier	ΔH_2^{abs}	0.36 eV	distinct templated-absorption closure
Absorption prefactor	A_{abs}	2.0	calibration; maximum preserving probability interpretation
Lattice parameter	a	2.867 Å	α -Fe material data
Atomic volume	Ω	11.80 Å ³	α -Fe material data; consistent with $a^3/2$
Core cutoff	δ	4.0 Å	dislocation-core model; Dudarev et al. (2008)
Softening temperature	T_c	1185 K	bcc-fcc transition/elastic softening; Dudarev et al. (2008)
111 prelog factor	$\hat{F}_{111}$	0.28 eV/Å	digitized/selected from anisotropic-elastic data
100 zero- T factor	$\hat{F}_{100}^0$	0.5182 eV/Å	derived from Eq. (15), not independent
111 core terms	$F_{\delta}^{111}, F_c^{111}$	0.345, 0.46 eV/Å	Dudarev et al. (2008)
100 core terms	$F_{\delta}^{100}, F_c^{100}$	0.387, 0.33 eV/Å	Dudarev et al. (2008)
Crossover temperature	T^*	450 °C	calibration condition in Eq. (15)
Reference size	n_{ref}	30	calibration condition; supplied analysis value

All three conserve SIA content by stoichiometry. The unary channel conserves loop number; the two binary channels reduce loop number by one per event. The clear separation of thermodynamic input, encounter kinetics, and calibrated branching factors is essential for mechanism-ablation studies and uncertainty quantification.

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